

Introduction:

This time I move forward with Advance Analysis using SQL to perform time-series analysis, Cumulative analysis, Product Performance analysis etc. I also designed a customers and products report which consolidates key metrics and KPIs for each customer and products.

1. Sales Performance Over Years

	year	no_of_days	total_sales	total_customers	total_quantity_sold
1	2010	3	43419	14	14
2	2011	365	7075088	2216	2216
3	2012	366	5842231	3255	3397
4	2013	365	16344878	17427	52807
5	2014	28	45642	834	1970

2. Running Total Sales and Moving Average over years

	year	total_sales_per_year	running_total	moving_avg_sales
1	2010	43419	43419	3101
2	2011	7075088	7118507	3146
3	2012	5842231	12960738	2670
4	2013	16344878	29305616	2080
5	2014	45642	29351258	1668

3. Yearly Performance of Products

(only 6 products performance are shown here)

	year	product_name	cy_sale	avg_sale	py_sale	cy_vs_avg	cy_vs_py
1	2012	All-Purpose Bike Stand	159	13197	0	Below Avg	No change
2	2013	All-Purpose Bike Stand	37683	13197	159	Above Avg	Increase
3	2014	All-Purpose Bike Stand	1749	13197	37683	Below Avg	Decrease
4	2012	AWC Logo Cap	72	6570	0	Below Avg	No change
5	2013	AWC Logo Cap	18891	6570	72	Above Avg	Increase
6	2014	AWC Logo Cap	747	6570	18891	Below Avg	Decrease
7	2013	Bike Wash - Dissolver	6960	3636	0	Above Avg	No change
8	2014	Bike Wash - Dissolver	312	3636	6960	Below Avg	Decrease
9	2013	Classic Vest- L	11968	6240	0	Above Avg	No change
10	2014	Classic Vest- L	512	6240	11968	Below Avg	Decrease
11	2013	Classic Vest- M	11840	6368	0	Above Avg	No change
12	2014	Classic Vest- M	896	6368	11840	Below Avg	Decrease
13	2012	Classic Vest- S	64	3648	0	Below Avg	No change
14	2013	Classic Vest- S	10368	3648	64	Above Avg	Increase
15	2014	Classic Vest- S	512	3648	10368	Below Avg	Decrease

4. Categories Revenue Contribution

	category	category_sales	total_sales	sales_contribution_percentage
1	Bikes	28316272	29356250	96.46%
2	Accessories	700262	29356250	2.39%
3	Clothing	339716	29356250	1.16%

5. No of products in different cost ranges

	cost_ranges	no of products
1	0-500	211
2	501-1500	63
3	Above 1500	21

6. Customers By Segment

	customer_segment	no_of_customers
1	New	14631
2	VIP	1655
3	Regular	2198

7. Customer Report (Only 15 customers shown here)

	customer_key	customer_name	age	age_group	customer_segment	last_order_date	recency	total_orders	total_sales	total_profit	total_quantity	lifespan	average_order_value	average_monthly_spend
1	1	Jon Yang	55	Above 50	VIP	2013-05-03	159	3	8249	3513	8	28	2749	294
2	2	Eugene Huang	50	40-50	VIP	2013-12-10	152	3	6384	2795	11	35	2128	182
3	3	Ruben Torres	55	Above 50	VIP	2013-02-23	162	3	8114	3455	4	25	2704	324
4	4	Christy Zhu	53	Above 50	VIP	2013-05-10	159	3	8139	3466	9	29	2713	280
5	5	Elizabeth Johnson	47	40-50	VIP	2013-05-01	159	3	8196	3502	6	28	2732	292
6	6	Julio Ruiz	50	40-50	VIP	2013-05-02	159	3	8121	3459	6	29	2707	280
7	7	Janet Alvarez	50	40-50	VIP	2013-05-14	159	3	8119	3453	5	28	2706	289
8	8	Marco Mehta	57	Above 50	VIP	2013-03-19	161	3	8211	3516	8	26	2737	315
9	9	Rob Verhoff	51	Above 50	VIP	2013-03-02	161	3	8106	3454	7	26	2702	311
10	10	Shannon Carlson	57	Above 50	VIP	2013-05-09	159	3	8091	3445	5	28	2697	288
11	11	Jacquelyn Suarez	57	Above 50	VIP	2013-05-23	159	3	8088	3435	4	28	2696	288
12	12	Curtis Lu	57	Above 50	VIP	2013-03-19	161	3	8133	3445	4	27	2711	301
13	13	Lauren Walker	47	40-50	New	2013-10-15	154	2	81	51	5	7	40	11
14	14	Ian Jenkins	47	40-50	New	2014-01-21	151	2	114	72	5	9	57	12
15	15	Sydney Bennett	53	Above 50	New	2013-04-30	160	2	138	87	6	1	69	138

8. Product Report (Only 15 products shown here)

	product_key	product_name	category	subcategory	cost	product_segment	total_orders	total_sales	total_quantity	total_profit	total_customers	lifespan	recency	average_order_revenue	average_monthly_revenue
1	3	Mountain-100 Black- 38	Bikes	Mountain Bikes	1898	High-Performers	49	165375	49	72373	285084	11	176	3375	15034
2	4	Mountain-100 Black- 42	Bikes	Mountain Bikes	1898	High-Performers	45	151875	45	66465	289628	11	176	3375	13806
3	5	Mountain-100 Black- 44	Bikes	Mountain Bikes	1898	High-Performers	60	202500	60	88620	438349	11	176	3375	18409
4	6	Mountain-100 Black- 48	Bikes	Mountain Bikes	1898	High-Performers	57	192375	57	84189	283559	12	176	3375	16031
5	7	Mountain-100 Silver- 38	Bikes	Mountain Bikes	1912	High-Performers	58	197200	58	86304	426121	12	176	3400	16433
6	8	Mountain-100 Silver- 42	Bikes	Mountain Bikes	1912	High-Performers	42	142800	42	62496	224121	11	176	3400	12981
7	9	Mountain-100 Silver- 44	Bikes	Mountain Bikes	1912	High-Performers	49	166600	49	72912	223248	12	176	3400	13883
8	10	Mountain-100 Silver- 48	Bikes	Mountain Bikes	1912	High-Performers	36	122400	36	53568	169132	11	176	3400	11127
9	16	Road-150 Red- 44	Bikes	Road Bikes	2171	High-Performers	281	1005418	281	395367	2203559	12	176	3578	83784
10	17	Road-150 Red- 48	Bikes	Road Bikes	2171	High-Performers	337	1205786	337	474159	2785558	12	176	3578	100482
11	18	Road-150 Red- 52	Bikes	Road Bikes	2171	High-Performers	302	1080556	302	424914	2462587	12	176	3578	90046
12	19	Road-150 Red- 56	Bikes	Road Bikes	2171	High-Performers	295	1055510	295	415065	2417991	12	176	3578	87959
13	20	Road-150 Red- 62	Bikes	Road Bikes	2171	High-Performers	336	1202208	336	472752	2783406	12	176	3578	100184
14	36	Road-650 Black- 44	Bikes	Road Bikes	487	Mid-Range	63	47565	63	16884	611876	23	164	755	2068
15	37	Road-650 Black- 48	Bikes	Road Bikes	487	Mid-Range	60	45552	60	16332	533255	21	164	759	2169